IMPLEMENTATION OF IP REPUTATION ENRICHMENT BOT USING N8N

I developed an **IP Reputation Enrichment Bot** using the n8n automation platform to analyse IPv4 addresses and generate threat intelligence reports through Telegram.

Initially, I created an n8n Cloud account and designed a workflow named **IP Reputation Enrichment Bot**. I created a Telegram bot using BotFather and connected it to n8n using secure Telegram API credentials.

Next, I configured the **Telegram Trigger** node to receive commands such as /ioc 8.8.8.8. The **Extract IP** node extracts the IP address from the message, and an IF node checks whether an IP address was provided. Invalid inputs are handled by sending a usage-help message.

I then integrated two threat intelligence services, **AbuseIPDB and VirusTotal**, using HTTP Request nodes. AbuseIPDB provides the abuse confidence score and reporting details, while VirusTotal provides malicious detection counts and reputation information. Both API keys were stored securely using n8n credentials.

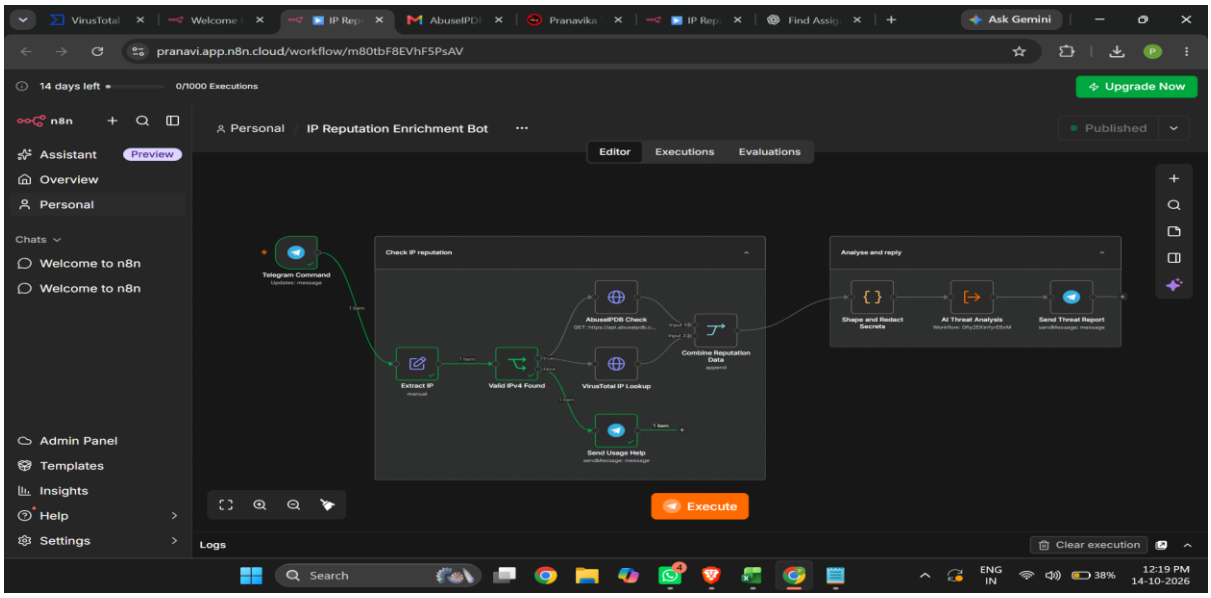
The results from both services were combined using a **Merge** node. A JavaScript Code node named **Shape and Redact Secrets** extracted relevant information and masked common sensitive-data patterns before passing the results for AI analysis.

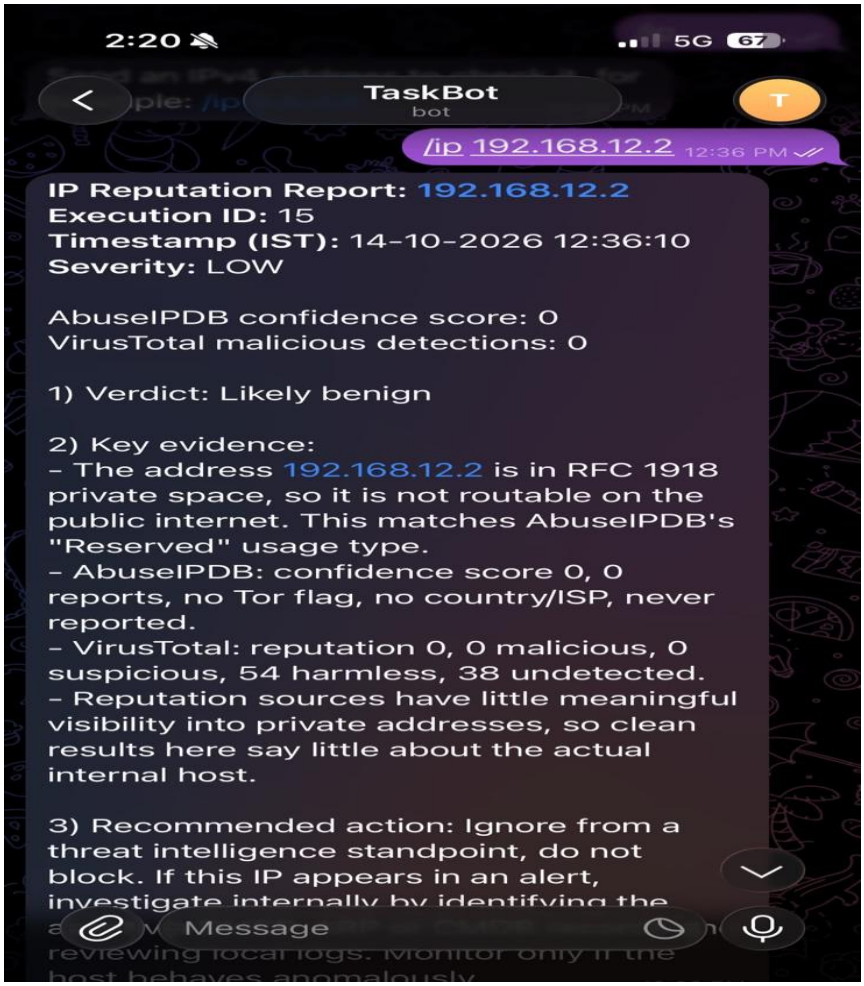
For the F6 requirement, I created a separate **AI Analysis sub-workflow** using an Anthropic Claude model. The main workflow calls this sub-workflow through the Execute Workflow node to generate a threat verdict, supporting evidence, and recommended action.

Finally, I configured the **Send Threat Report** node to return the results through Telegram, including the IP address, execution ID, IST timestamp, severity level, AbuseIPDB score, VirusTotal detections, AI verdict, and recommended action.

I tested the workflow using 8.8.8.8 and 1.1.1.1 and successfully received threat intelligence reports. I also tested invalid input using /ioc hello, which produced a usage-help response. After verifying the workflow, I published both the main workflow and AI sub-workflow and exported their JSON files.

Thus, I successfully implemented an automated IP reputation analysis system using **n8n, Telegram, AbuseIPDB, VirusTotal, JavaScript, and AI-based threat assessment**.


pranavi.app.n8n.cloud/workflow/m80tbF8EVhF5PsAV
14 days left
0/1000 Executions
Upgrade Now
n8n
Personal
IP Reputation Enrichment Bot
Published
Assistant
Preview
Overview
Personal
Chats
Welcome to n8n
Welcome to n8n
Admin Panel
Templates
Insights
Help
Settings
Editor
Executions
Evaluations
Telegram Command
Check IP reputation
AbuseIPDB Check
Extract IP
Valid IPv4 Found
VirusTotal IP Lookup
Combine Reputation Data
Send Usage Help
Analyse and reply
Shape and Redact Secrets
AI Threat Analysis
Send Threat Report
Execute
Logs
Clear execution
Search
ENG
IN
38%
12:19 PM
14-10-2026


2:20
5G
67
TaskBot
bot
/ip 192.168.12.2 12:36 PM
IP Reputation Report: 192.168.12.2
Execution ID: 15
Timestamp (IST): 14-10-2026 12:36:10
Severity: LOW
AbuseIPDB confidence score: 0
VirusTotal malicious detections: 0
1) Verdict: Likely benign
2) Key evidence:
- The address 192.168.12.2 is in RFC 1918 private space, so it is not routable on the public internet. This matches AbuseIPDB's "Reserved" usage type.
- AbuseIPDB: confidence score 0, 0 reports, no Tor flag, no country/ISP, never reported.
- VirusTotal: reputation 0, 0 malicious, 0 suspicious, 54 harmless, 38 undetected.
- Reputation sources have little meaningful visibility into private addresses, so clean results here say little about the actual internal host.
3) Recommended action: Ignore from a threat intelligence standpoint, do not block. If this IP appears in an alert, investigate internally by identifying the
Message